Institut de Mathématiques de Toulouse, INSA Toulouse

Kernel methods, Support Vector Machine Support Vector Regression

Machine Learning
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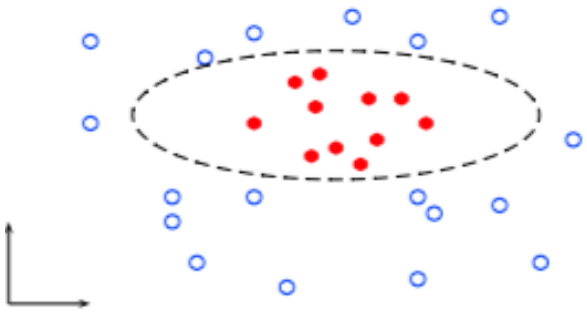
Béatrice Laurent

Outline

- Introduction
- The kernel trick
- Minimization of the convexified empirical risk
- Support Vector Regression
- Kernel regularized least square
- Conclusion

Introduction

- In the previous course, we have studied linear SVM.
- The assumption was that the training set is nearly linearly separable.
- In most cases, this assumption is not realistic.



- In this case, a linear SVM leads to bad performances.

Introduction

- We can make the classification procedure more flexible by enlarging the feature space and sending the entries $\{x_i, i = 1 \dots n\}$ in an Hilbert space $\mathcal{H}$, with high or possibly infinite dimension, via a function ϕ, and we apply a linear SVM procedure on the new training set $\{(\phi(x_i), y_i), i = 1 \dots n\}$.
- The space $\mathcal{H}$ is called the **feature space**. This idea is due to Boser, Guyon, Vapnik (1992).
- In the previous example, setting $\phi(x) = (x_1^2, x_2^2, x_1 x_2)$, the training set becomes linearly separable in $\mathbb{R}^3$, and a linear SVM is appropriate.

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The kernel trick

- A natural question arises : how can we choose $\mathcal{H}$ and ϕ ? In fact, we do not choose $\mathcal{H}$ and ϕ but a *kernel* .
- The classification rule is

$$\hat{f}(x) = \mathbb{1}_{\sum y_i \alpha_i^* \langle \phi(x_i), \phi(x) \rangle + b^* \geq 0} - \mathbb{1}_{\sum y_i \alpha_i^* \langle \phi(x_i), \phi(x) \rangle + b^* < 0},$$

  where the α_i^*'s are the solutions of the dual problem in the feature space $\mathcal{H}$:
- Maximizing $\theta(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j \langle \phi(x_i), \phi(x_j) \rangle$ s. t. $\sum_{i=1}^n \alpha_i y_i = 0$ and $0 \leq \alpha_i \leq C \ \forall i$.

- It is important to notice that the final classification rule in the feature space depends on ϕ only through scalar products of the form $\langle \phi(x_i), \phi(x) \rangle$ or $\langle \phi(x_i), \phi(x_j) \rangle$.

- The only knowledge of the function k defined by $k(x, x') = \langle \phi(x), \phi(x') \rangle$ allows to define the SVM in the feature space $\mathcal{H}$ and to derive a classification rule in the space $\mathcal{X}$. The explicit computation of ϕ is not required.

Definition

A function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ such that $k(x, x') = \langle \phi(x), \phi(x') \rangle$ for a given function $\phi : \mathcal{X} \to \mathcal{H}$ is called a **kernel**.

- A kernel is generally more easy to compute than the function ϕ that returns values in a high dimensional space. For example, for $x = (x_1, x_2) \in \mathbb{R}^2$, $\phi(x) = (x_1^2, \sqrt{2}x_1x_2, x_2^2)$, and $k(x, x') = \langle x, x' \rangle^2$.
- Let us now give a property to ensure that a function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ defines a kernel.

Proposition

. —**Mercer condition** *If the function* $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ *is continuous, symmetric, and if for all finite subset* $\{x_1, \ldots, x_n\}$ *in* $\mathcal{X}$*, the matrix* $(k(x_i, x_j))_{1 \leq i,j \leq n}$ *is positive definite :*

$$\forall c_1, \ldots, c_n \in \mathbb{R}, \sum_{i,j=1}^{n} c_i c_j k(x_i, x_j) \geq 0,$$

then, there exists an Hilbert space $\mathcal{H}$ *and a function* $\phi : \mathcal{X} \to \mathcal{H}$ *such that* $k(x, x') = \langle \phi(x), \phi(x') \rangle_{\mathcal{H}}$*. The space* $\mathcal{H}$ *is called the* **Reproducing kernel Hilbert Space (RKHS)** *associated to* k*.*
We have :

1. *For all* $x \in \mathcal{X}$*,* $k(x, .) \in \mathcal{H}$ *where* $k(x, .) : y \mapsto k(x, y)$*.*
2. **Reproducing property** *:*

$$h(x) = \langle h, k(x, .) \rangle_{\mathcal{H}} \text{ for all } x \in \mathcal{X} \text{ and } h \in \mathcal{H}.$$

- Let us give some examples.
- The Mercer condition is often hard to verify but we know some classical examples of kernels that can be used.
- We assume that $\mathcal{X} = \mathbb{R}^p$.
  - p **degree polynomial kernel** : $k(x, x') = (1 + \langle x, x' \rangle)^p$
  - **Gaussian kernel (RBF)** : $k(x, x') = e^{-\frac{\|x - x'\|^2}{2\sigma^2}}$
    ϕ returns values in a infinite dimensional space.
  - **Laplacian kernel** : $k(x, x') = e^{-\frac{\|x - x'\|}{\sigma}}$.
  - **Sigmoid kernel** : $k(x, x') = \tanh(\kappa \langle x, x' \rangle + \theta)$

- We have seen some examples of kernels. One can construct new kernels by aggregating several kernels.
- For example let k_1 and k_2 be two kernels and f a function $\mathbb{R}^p \to \mathbb{R}$, $\phi : \mathbb{R}^p \to \mathbb{R}^{d'}$, B a positive definite matrix, P a polynomial with positive coefficients and $\lambda > 0$.
  The functions defined by $k(x, x') = k_1(x, x') + k_2(x, x')$, $\lambda k_1(x, x')$, $k_1(x, x')k_2(x, x')$, $k_1(\phi(x), \phi(x'))$, $x^T Bx'$, $P(k_1(x, x'))$, or $e^{k_1(x,x')}$ are still kernels.
- We have presented examples of kernels for the case where $\mathcal{X} = \mathbb{R}^p$ but a very interesting property is that kernels can be defined for very general input spaces, such as **sets, trees, graphs, texts, DNA sequences ...**

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Minimization of the convexified empirical risk

The ideal classification rule minimizes the risk $L(f) = \mathbb{P}(Y \neq f(X))$, the solution is the Bayes rule f^*.
The risk is replaced by the empirical risk to be minimized :

$$L_n(f) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{Y_i \neq f(X_i)}.$$

In order to avoid overfitting, the minimization is restricted to a set $\mathcal{F}$:

$$\hat{f} = \mathrm{argmin}_{f \in \mathcal{F}} L_n(f).$$

The risk of $\hat{f}$ can be decomposed in two terms :

$$0 \leq L(\hat{f}) - L(f^*) = \min_{f \in \mathcal{F}} L(f) - L(f^*) + L(\hat{f}) - \min_{f \in \mathcal{F}} L(f).$$

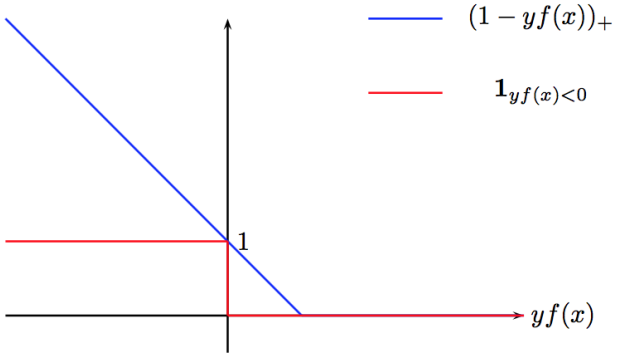
The first term $\min_{f \in \mathcal{F}} L(f) - L(f^*)$ is the approximation error, or bias term, the second term $L(\hat{f}) - \min_{f \in \mathcal{F}} L(f)$ is the stochastic error or variance term.

Enlarging the class $\mathcal{F}$ reduces the approximation error but increases the stochastic error.
The empirical risk minimization classifier cannot be used in practice because of its computational cost, indeed L_n is not convex.
This is the reason why we generally replace the empirical misclassification probability L_n by some convex surrogate, and we consider convex classes $\mathcal{F}$.
We consider a loss function ℓ, and we require the condition $\mathbb{1}_{z<0} \leq \ell(z)$, which will allow to give an upper bound for the misclassification probability ; indeed

$$\mathbb{E}(\mathbb{1}_{Yf(X)<0}) = \mathbb{P}(Y \neq f(X)) \leq \mathbb{E}(\ell(Yf(X))).$$

Classical convex losses ℓ are :

- the hinge loss $\ell(z) = (1 - z)_+$,
- the exponential loss $\ell(z) = \exp(-z)$,
- the logit loss $\ell(z) = \log_2(1 + \exp(-z))$.



Let us show that SVM are solutions of the minimization of the convexified and penalized empirical risk.

For the sake of simplicity, we consider the linear case.
We first notice that the following optimization problem :

Minimizing $\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i$ s. t. $\begin{cases} y_i\left(\langle w, x_i\rangle + b\right) \geq 1 - \xi_i \ \forall\ i \\ \xi_i \geq 0 \end{cases}$

is equivalent to minimize

$$\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\left(1 - y_i\left(\langle w, x_i\rangle + b\right)\right)_+ ,$$

or equivalently

$$\frac{1}{n}\sum_{i}^{n}\left(1 - y_i\left(\langle w, x_i\rangle + b\right)\right)_+ + \frac{1}{2Cn}\|w\|^2.$$

Hence, SVM are solutions of the minimization of the convexified empirical risk with the hinge loss ℓ plus a penalty term.
Indeed, SVM are solutions of

$$\mathrm{argmin}_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \ell(y_i f(x_i)) + \mathrm{pen}(f),$$

where

$$\mathcal{F} = \{\langle w, x \rangle + b, w \in \mathbb{R}^p, b \in \mathbb{R}\}$$

and

$$\forall f \in \mathcal{F}, \mathrm{pen}(f) = \frac{1}{2Cn} \|w\|^2.$$

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Support Vector Regression

- Kernel methods can also be used for regression function estimation.
- Suppose we have a training sample $\{(x_1, y_1), \ldots, (x_n, y_n)\} \in (\mathcal{X} \times \mathbb{R})^n$, where $\mathcal{X}$ denotes the space of the inputs (for example $\mathcal{X} = \mathbb{R}^p$).
- The ε support vector regression, introduced by Vapnik (1995) aims to find a function f such that for all i, the deviation between $f(x_i)$ and y_i is at most ε, and such that, at the same time, f is as "flat" as possible.
- Let us first consider the case of linear predictors :

$$f(x) = \langle w, x \rangle + b, \text{ with } x \in \mathcal{X}, b \in \mathbb{R}\,.$$

- Flatness means here that $\|w\|$ is small.

Support Vector Regression

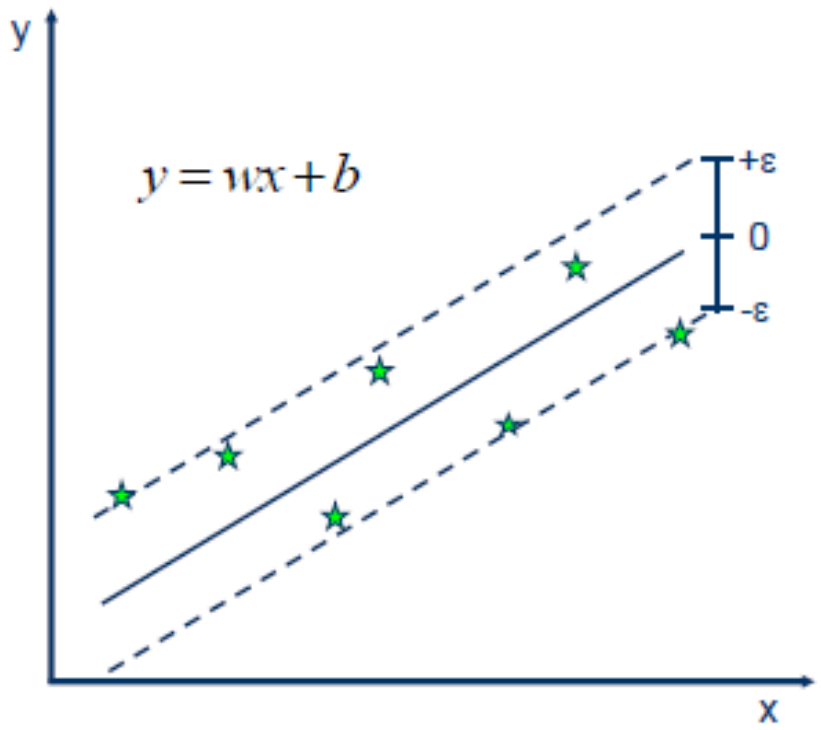
- This leads to the convex optimization problem :

$$\text{Minimize } \frac{1}{2}\|w\|^2$$
$$\text{under the constraints, for all } i$$

$$\begin{cases} y_i - (\langle w, x_i \rangle + b) \leq \varepsilon \\ -y_i + (\langle w, x_i \rangle + b) \leq \varepsilon \end{cases}$$

- Note that here, we do not care of errors less than ε, but we do not accept errors greater than ε.
- The tacit assumption is that the above problem admits a solution.

Support Vector Regression



- Solution:

$$\min \frac{1}{2}\|w\|^2$$

- Constraints:

$$y_i - wx_i - b \leq \varepsilon$$

$$wx_i + b - y_i \leq \varepsilon$$

Support Vector Regression

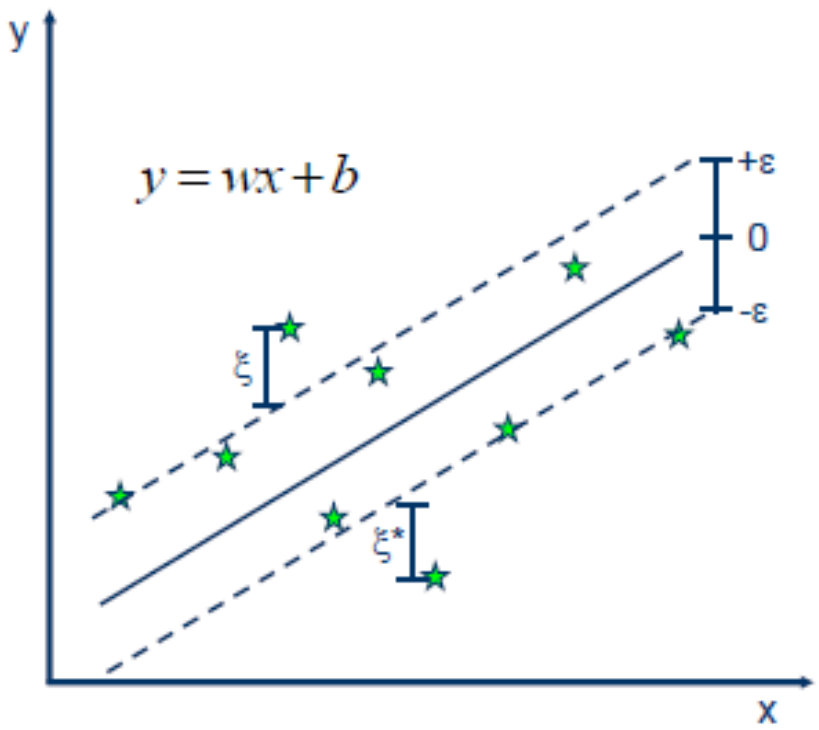
- To be more general, we want to allow some errors.
- Like for the classification problem, we introduce slack variables ξ_i, ξ_i' to overcome possible unfeasible constraints in the previous optimization problem.
- This leads to the following formulation :

$$\text{Minimize } \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i') \tag{1}$$

under the constraints, for all i

$$\begin{cases} y_i - (\langle w, x_i\rangle + b) \leq \varepsilon + \xi_i \\ -y_i + (\langle w, x_i\rangle + b) \leq \varepsilon + \xi_i' \\ \xi_i, \xi_i' \geq 0 \end{cases}$$

Support Vector Regression



- Minimize:

$$\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{N}\left(\xi_i + \xi_i^*\right)$$

- Constraints:

$$y_i - wx_i - b \le \varepsilon + \xi_i$$

$$wx_i + b - y_i \le \varepsilon + \xi_i^*$$

$$\xi_i, \xi_i^* \ge 0$$

Exercise : Prove that this optimization problem is equivalent to the minimization of

$$\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} \ell_\varepsilon(y_i, f(x_i)),$$

where ℓ_ε is the so-called ε-insensitive loss function defined by

$$\begin{aligned} \ell_\varepsilon(y, y') &= 0 \qquad\qquad \text{if } |y - y'| \leq \varepsilon \\ &= |y - y'| - \varepsilon \text{ otherwise .} \end{aligned}$$

- In most cases, the optimization problem (1) can be solved more easily in its dual formulation.
- Moreover, like for classification problems, the dual formulation allows to extend easily the support vector regression to nonlinear functions.
- The Lagrangian is

$$L(w, b, \xi, \xi', \eta, \eta', \alpha, \alpha') = \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i') - \sum_{i=1}^{n}(\eta_i\xi_i + \eta_i'\xi_i')$$
$$- \sum_{i=1}^{n}\alpha_i(\varepsilon + \xi_i - y_i + \langle w, x_i\rangle + b) - \sum_{i=1}^{n}\alpha_i'(\varepsilon + \xi_i' + y_i - \langle w, x_i\rangle - b),$$

with $\eta, \eta', \alpha, \alpha'$ the Lagrange multipliers, $\eta_i, \eta_i', \alpha_i, \alpha_i' \geq 0$.

- The cancellation of the partial derivatives with respect to the primal variables $\frac{\partial L}{\partial w}(w, b, \xi, \xi', \eta, \eta', \alpha, \alpha')$, $\frac{\partial L}{\partial b}(w, b, \xi, \xi', \eta, \eta', \alpha, \alpha')$ and $\frac{\partial L}{\partial \xi_i^{(\prime)}}(w, b, \xi, \xi', \eta, \eta', \alpha, \alpha')$ leads to the following dual problem.
- **Dual problem.** Show that the dual problem can be formulated as follows :

$$\begin{aligned}
\text{Maximize} \quad & -\frac{1}{2}\sum_{i,j=1}^{n}(\alpha_i - \alpha_i')(\alpha_j - \alpha_j')\langle x_i, x_j\rangle \\
& -\varepsilon\sum_{i=1}^{n}(\alpha_i + \alpha_i') + \sum_{i=1}^{n} y_i(\alpha_i - \alpha_i') \\
\text{subject to} \quad & \sum_{i=1}^{n}(\alpha_i - \alpha_i') = 0 \text{ and } 0 \leq \alpha_i, \alpha_i' \leq C \ \forall i.
\end{aligned}$$

Karush-Kuhn-Tucker conditions :

- $\alpha_i^*(\varepsilon + \xi_i^* - y_i + \langle w^*, x_i \rangle + b^*) = 0$
- $(\alpha_i')^*(\varepsilon + (\xi_i')^* + y_i - \langle w^*, x_i \rangle - b^*) = 0$
- $\xi_i^*(C - \alpha_i^*) = 0$, $(\xi_i')^*(C - (\alpha_i')^*) = 0$

Exercise :

Draw a picture to show the support vectors for the regression problem.

The support vector regression function is :

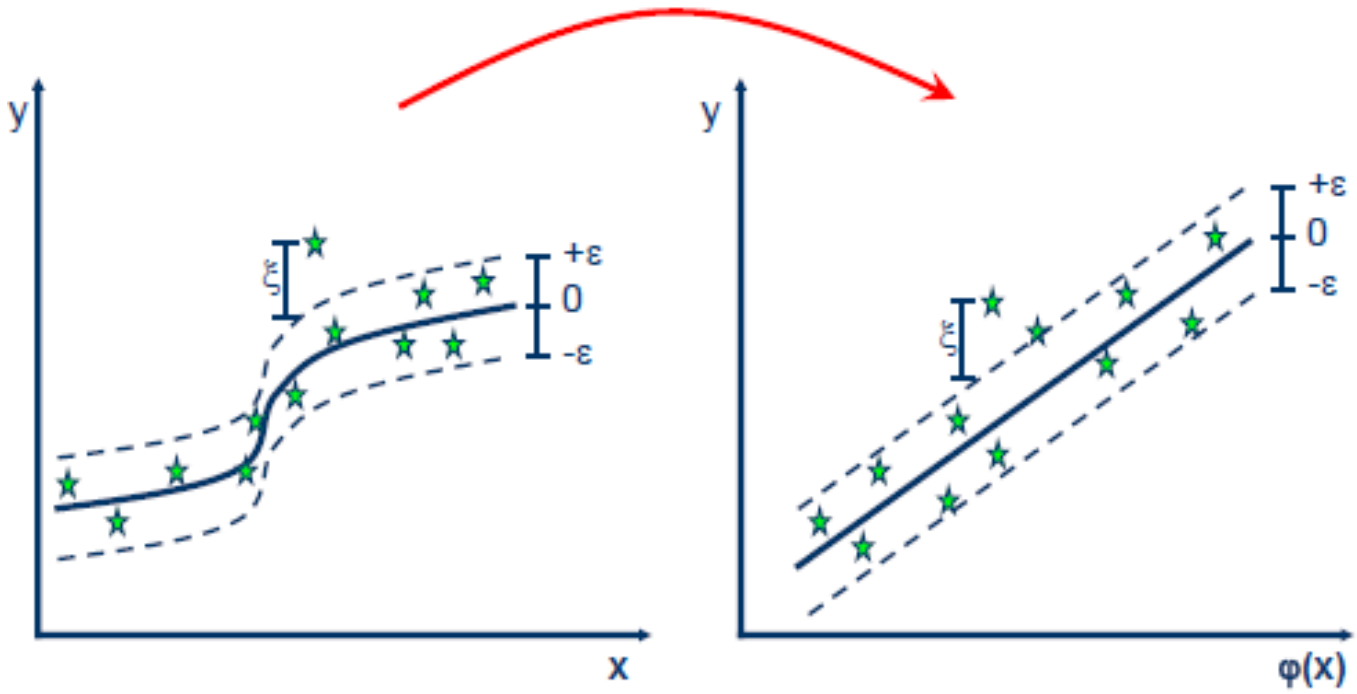
$$\begin{aligned} f^*(x) &= \langle w^*, x\rangle + b^* \\ &= \langle \sum_i (\alpha_i^* - \alpha_i'^*) x_i, x\rangle + b^* \\ &= \sum_i (\alpha_i^* - \alpha_i'^*)\langle x_i, x\rangle + b^*. \end{aligned}$$

As previously, only the scalar product $\langle x_i, x\rangle$ are involved in the solution, allowing easily to extend to nonlinear regression functions using kernels, leading to the predictor :

$$f^*(x) = \sum_{i\ sv} (\alpha_i^* - \alpha_i'^*)\langle \Phi(x_i), \Phi(x)\rangle + b^*$$

$$f^*(x) = \sum_{i\ sv} (\alpha_i^* - \alpha_i'^*) k(x_i, x) + b^*.$$

Support Vector Regression



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Kernel regularized least square

- We present here another regression method based on kernels : the Kernel regularized least Square procedure.
- It is based on a penalized least square criterion.
- Let $(\boldsymbol{X_i}, Y_i)_{1 \leq i \leq n}$ the observations, with $\boldsymbol{X_i} \in \mathbb{R}^p$, $Y_i \in \mathbb{R}$. We consider a positive definite kernel k defined on $\mathbb{R}^p$:

$$k(\boldsymbol{x}, \boldsymbol{y}) = k(\boldsymbol{y}, \boldsymbol{x}); \quad \sum_{i,j=1}^{n} c_i c_j k(\boldsymbol{X_i}, \boldsymbol{X_j}) \geq 0.$$

- We are looking for a predictor of the form

$$f(\boldsymbol{x}) = \sum_{i=1}^{n} c_j k(\boldsymbol{X_j}, \boldsymbol{x}), \ \boldsymbol{c} \in \mathbb{R}^n.$$

- Let us denote by $\boldsymbol{K}$ the matrix defined by $\boldsymbol{K}_{i,j} = k(\boldsymbol{X_i}, \boldsymbol{X_j})$.
- The KRLS method consists in minimizing for f on the form defined above the penalized least square criterion

$$\sum_{i=1}^{n} (Y_i - f(\boldsymbol{X_i}))^2 + \lambda \|f\|_{\boldsymbol{K}}^2,$$

where

$$\|f\|_{\boldsymbol{K}}^2 = \sum_{i,j=1}^{n} c_i c_j k(\boldsymbol{X_i}, \boldsymbol{X_j}).$$

- Equivalently, we minimize for $c \in \mathbb{R}^n$ the criterion

$$\|\boldsymbol{Y} - \boldsymbol{K}\boldsymbol{c}\|^2 + \lambda \boldsymbol{c}'\boldsymbol{K}\boldsymbol{c}.$$

If $\boldsymbol{K}$ is invertible, there exists an explicit solution

$$\hat{\boldsymbol{c}} = (\boldsymbol{K} + \lambda I_n)^{-1} Y,$$

which leads to the predictor

$$\hat{f}(\boldsymbol{x}) = \sum_{j=1}^{n} \hat{c}_j k(\boldsymbol{X_j}, \boldsymbol{x}).$$

$$\hat{\boldsymbol{Y}} = \boldsymbol{K}\hat{\boldsymbol{c}}.$$

With a kernel corresponding to the scalar product, we recover a linear predictor

$$\boldsymbol{K} = \boldsymbol{X}\boldsymbol{X}', \hat{\boldsymbol{c}} = (\boldsymbol{X}\boldsymbol{X}' + \lambda I_n)^{-1} Y,$$

$$\hat{f}(\boldsymbol{x}) = \sum_{j=1}^{n} \hat{c}_j \langle \boldsymbol{X_j}, \boldsymbol{x} \rangle.$$

- For polynomial or Gaussian kernels for example, we obtain non linear predictors.
- As for SVM, an important interest of this method is the possibility to be generalized to complex predictors such as text, graphs, DNA sequences .. as soon as one can define a kernel function on such objects.

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Conclusion

- Using kernels allows to delinearize classification algorithms by mapping $\mathcal{X}$ in the RKHS $\mathcal{H}$ with the map $x \mapsto k(x, .)$. It provides nonlinear algorithms with almost the same computational properties as linear ones.
- SVM have nice theoretical properties, cf. Vapnik's theory for empirical risk minimization.
- The use of RKHS allows to apply to any set $\mathcal{X}$ (such as set of graphs, texts, DNA sequences ..) algorithms that are defined for vectors as soon as we can define a kernel $k(x, y)$ corresponding to some measure of similarity between two objects of $\mathcal{X}$.

Conclusion

- Important issues concern the choice of the kernel, and of the tuning parameters to define the SVM procedure.
- Note that SVM can also be used for multi-class classification problems : one can use pairwise SVM classification for each class against another class (one versus one) and predict the class for a new point by a majority vote. Another way is to use a one versus all method, in this case each class is classified agains all the others (one model per class) ; in this case, a new sample is assigned to the class with the largest output.
- Kernels methods are also used for non supervised classification (kernel PCA), and for anomaly detection (One-class SVM).

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